



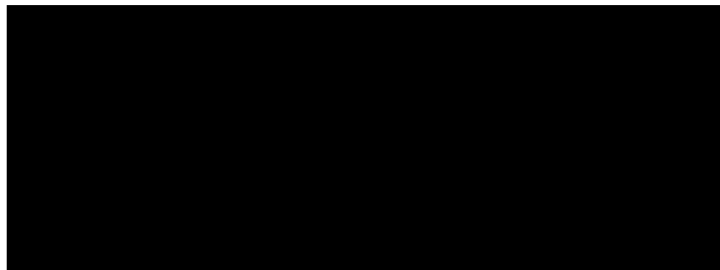
INDIVIDUAL ASSIGNMENT PART 1

TECHNOLOGY PARK MALAYSIA

CT015-3-2

DESIGN METHODS

APD2F2409SE



NAME : CHEOK SHAWN ANN



Table of Contents

Table Content of Figure.....	3
1.0 Design Model	4
1.1 Class Diagram	4
1.2 Sequence Diagram.....	5
1. UC1_Create Project based with Templates or Tools.....	5
2. UC3_Manage Task	6
3. UC5_Open Relevant File with URL Link.....	7
4. UC7_Solve Questions Using Notion AI	8
5. UC8_Searching Project and Task.....	9
6. UC10_View Trash History	10
7. UC11_Using Real Time Tracker	11
8. UC12_View Visualize Graph.....	12
1.3 State Machine Diagram.....	13
1. UC1_Create Project based with Templates or Tools.....	13
2. UC3_Manage Task	13
3. UC5_Open Relevant File with URL Link.....	14
4. UC7_Solve Questions Using Notion AI.....	14
5. UC8_Searching Project and Task.....	14
6. UC10_View Trash History	15
7. UC11_Using Real Time Tracker	16
8. UC12_View Visualize Graph.....	17
2.0 Additional Diagram.....	18
2.1 Object Diagram	18
1. UC1_Create Project based with Templates or Tools.....	18
2. UC12_View Visualize Graph.....	18
3.0 Conclusion.....	19
4.0 References	21

Table Content of Figure

Figure 1 Class Diagram.....	4
Figure 2 .UC1_ Create Project based with Templates or Tools (Sequence Diagram)	5
Figure 3 UC3_ Manage Task (Sequence Diagram).....	6
Figure 4 UC5_ Open Relevant File with URL Link (Sequence Diagram)	7
Figure 5 UC7_ Solve Questions Using Notion AI (Sequence Diagram)	8
Figure 6 UC8_ Searching Project and Tasks (Sequence Diagram)	9
Figure 7 UC10_ View Trash History (Sequence Diagram).....	10
Figure 8 UC11_ Using Real Time Tracker (Sequence Diagram).....	11
Figure 9 UC12_ View Visualize Graph (Sequence Diagram)	12
Figure 10 UC1_ Create Project based with Templates or Tools (State Machine Diagram)	13
Figure 11 UC3_ Manage Task (State Machine Diagram).....	13
Figure 12 UC5_ Open Relevant File with URL Link (State Machine Diagram)	14
Figure 13 UC7_ Solve Questions Using Notion AI (State Machine Diagram)	14
Figure 14 UC8_ Searching Project and Task (State Machine Diagram)	14
Figure 15 UC10_ View Trash History (State Machine Diagram).....	15
Figure 16 UC11_ Using Real Time Tracker (State Machine Diagram).....	16
Figure 17 UC12_ View Visualize Graph (State Machine Diagram)	17
Figure 18 UC1_ Create Project based with Templates or Tools (Objective Diagram).....	18
Figure 19 UC12_ View Visualize Graph (Objective Diagram).....	18

1.0 Design Model

1.1 Class Diagram

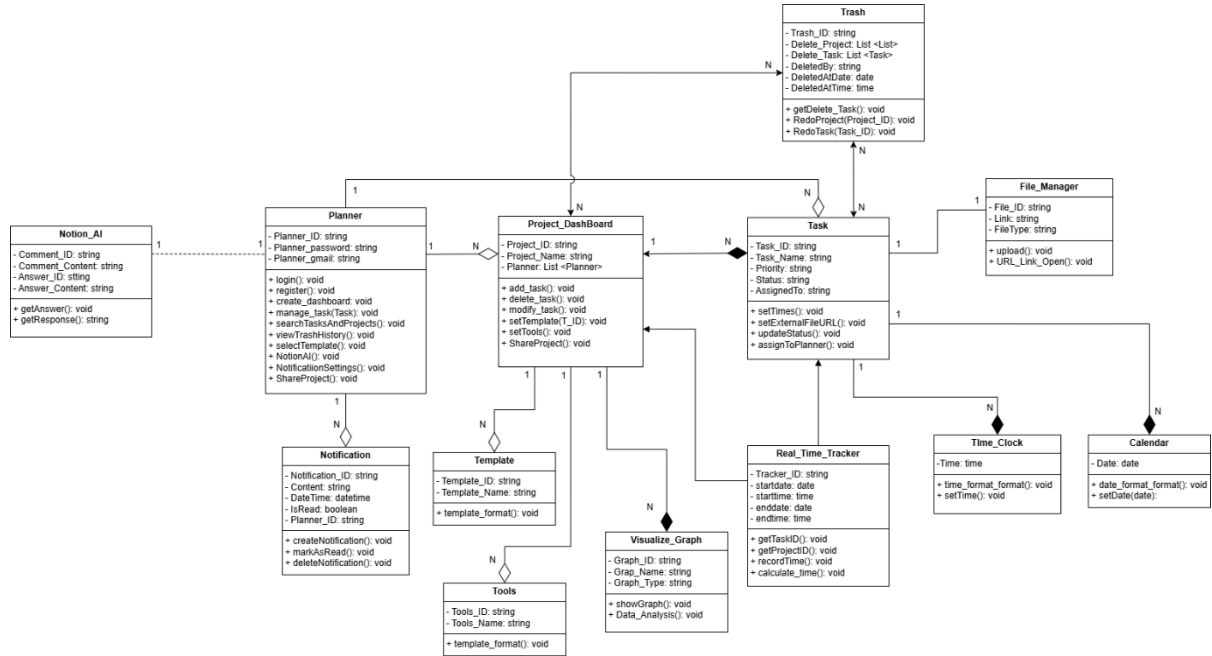


Figure 1 Class Diagram

1.2 Sequence Diagram

1. UC1_Create Project based with Templates or Tools

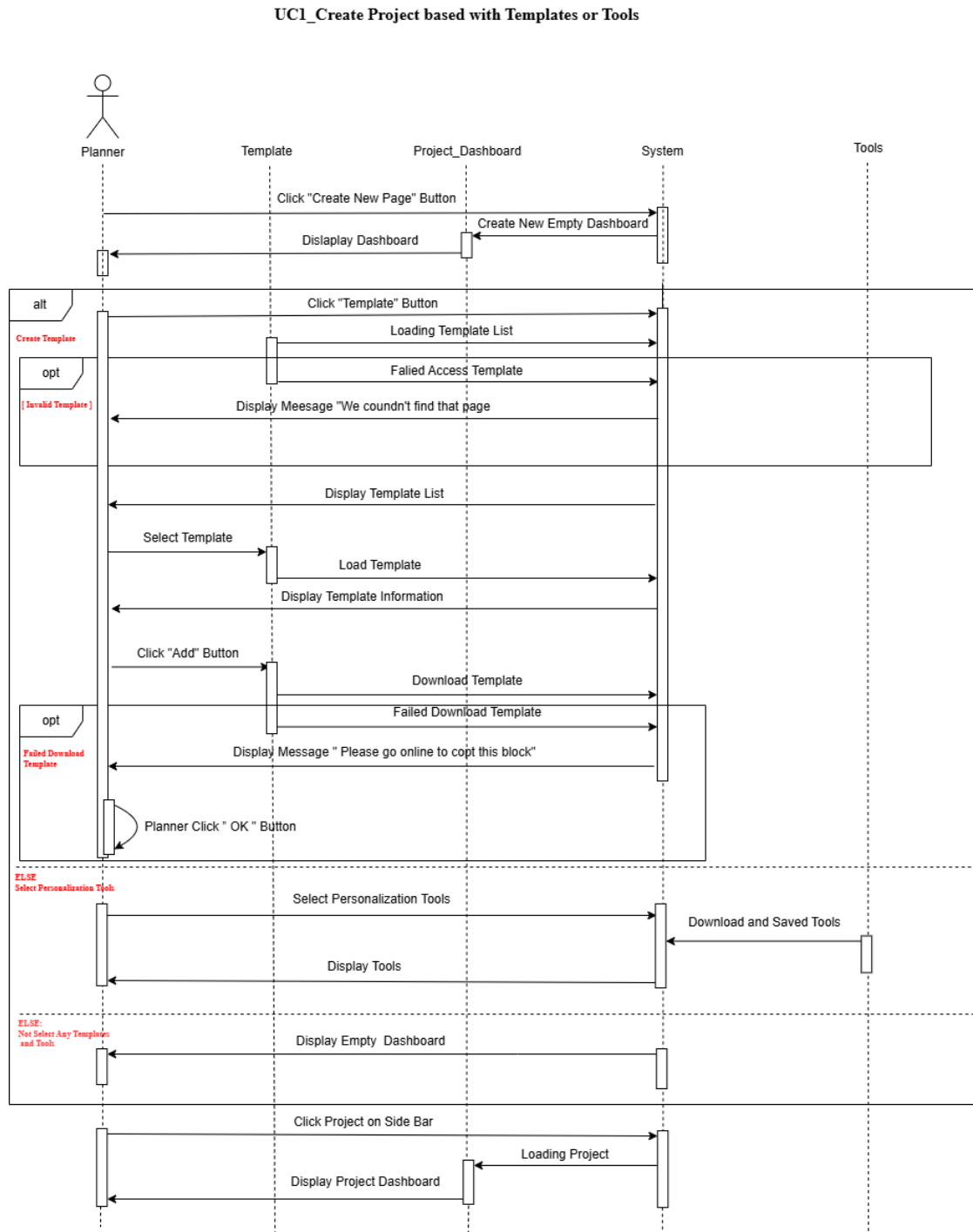


Figure 2 .UC1_Create Project based with Templates or Tools (Sequence Diagram)

2. UC3_Manage Task

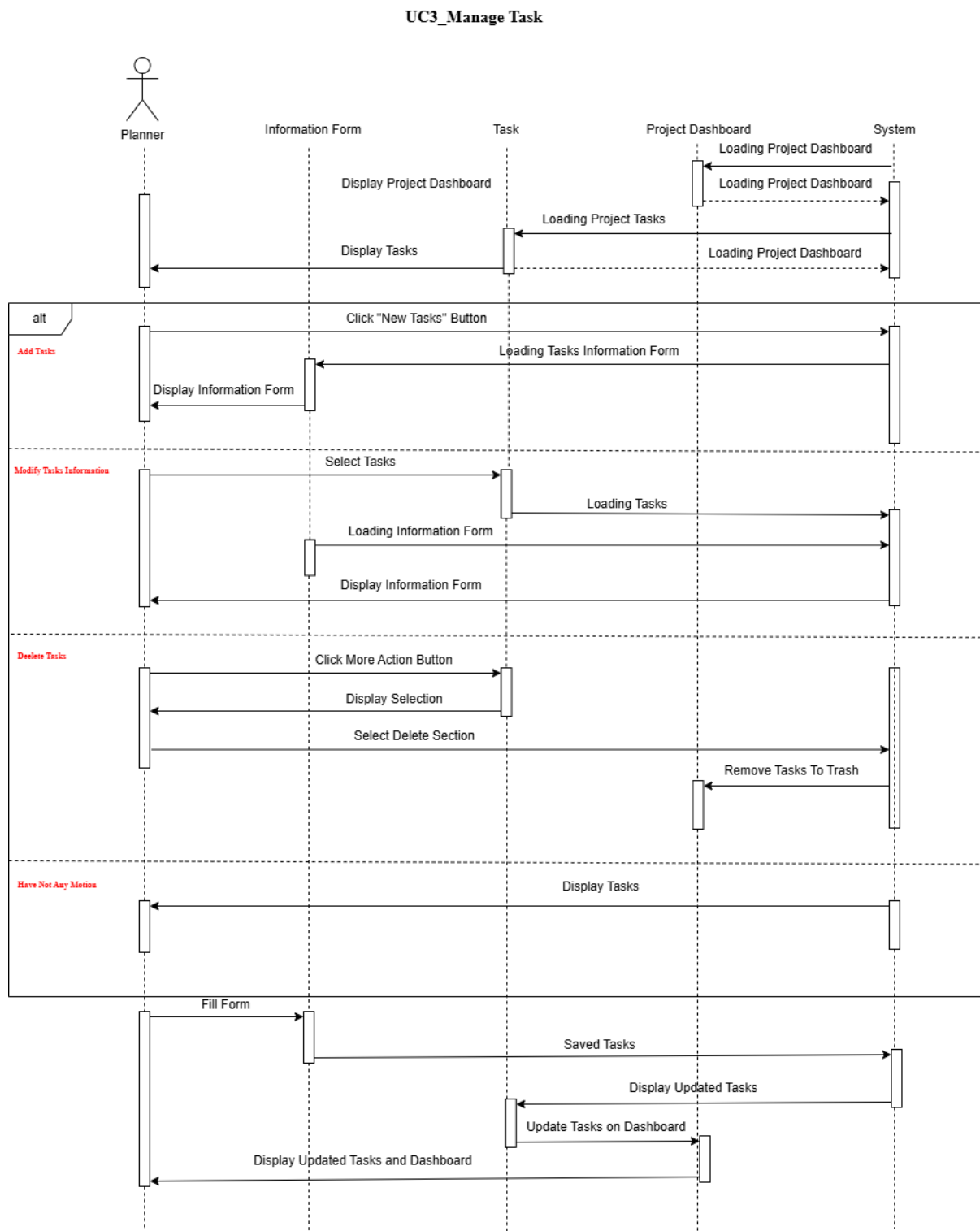


Figure 3 UC3_Manage Task (Sequence Diagram)

3. UC5_Open Relevant File with URL Link

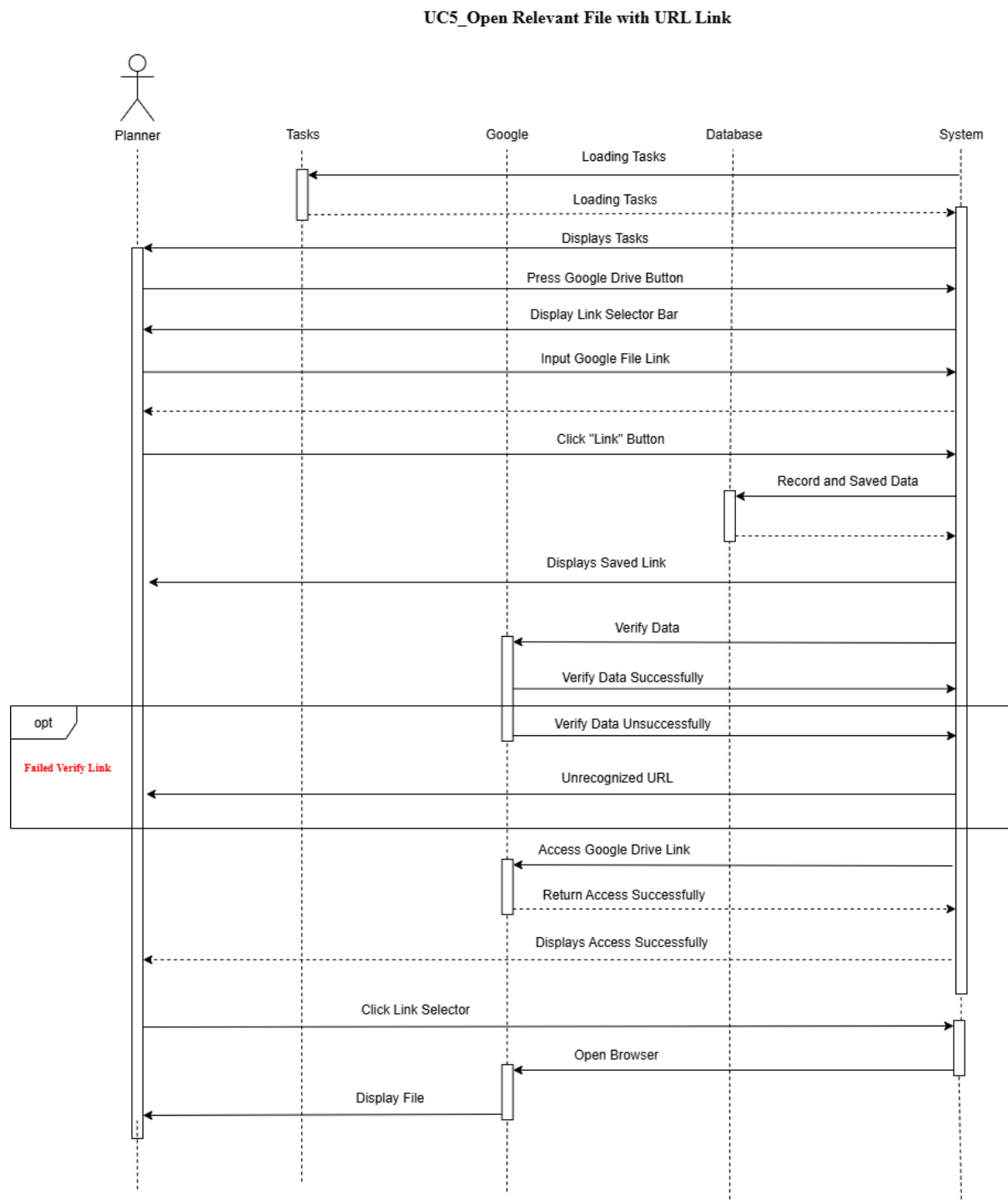


Figure 4 UC5_Open Relevant File with URL Link (Sequence Diagram)

4. UC7_Solve Questions Using Notion AI

UC7_Solve Questions Using Notion AI

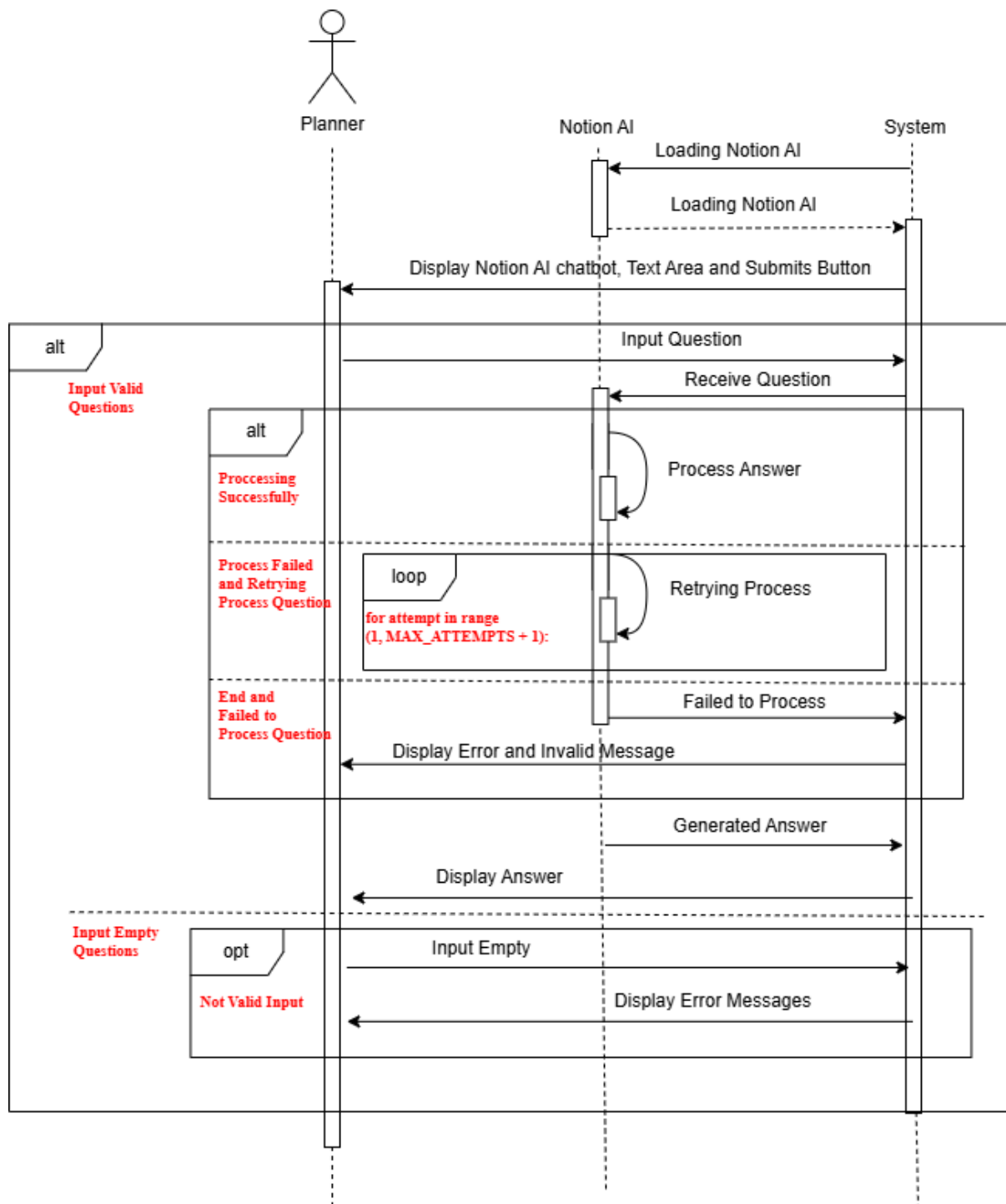


Figure 5 UC7_Solve Questions Using Notion AI (Sequence Diagram)

5. UC8_Searching Project and Task

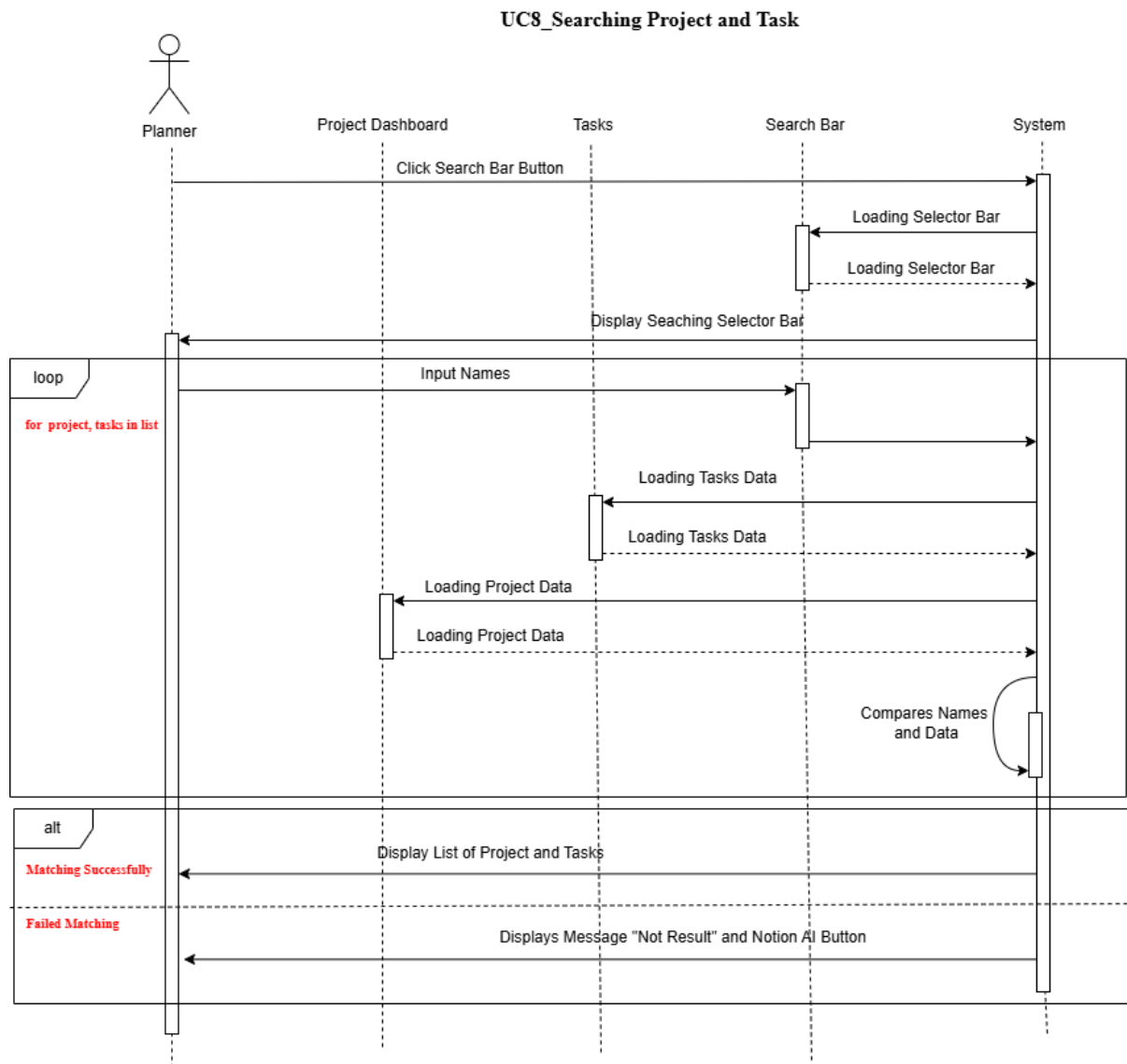


Figure 6 UC8_Searching Project and Tasks (Sequence Diagram)

6. UC10_View Trash History

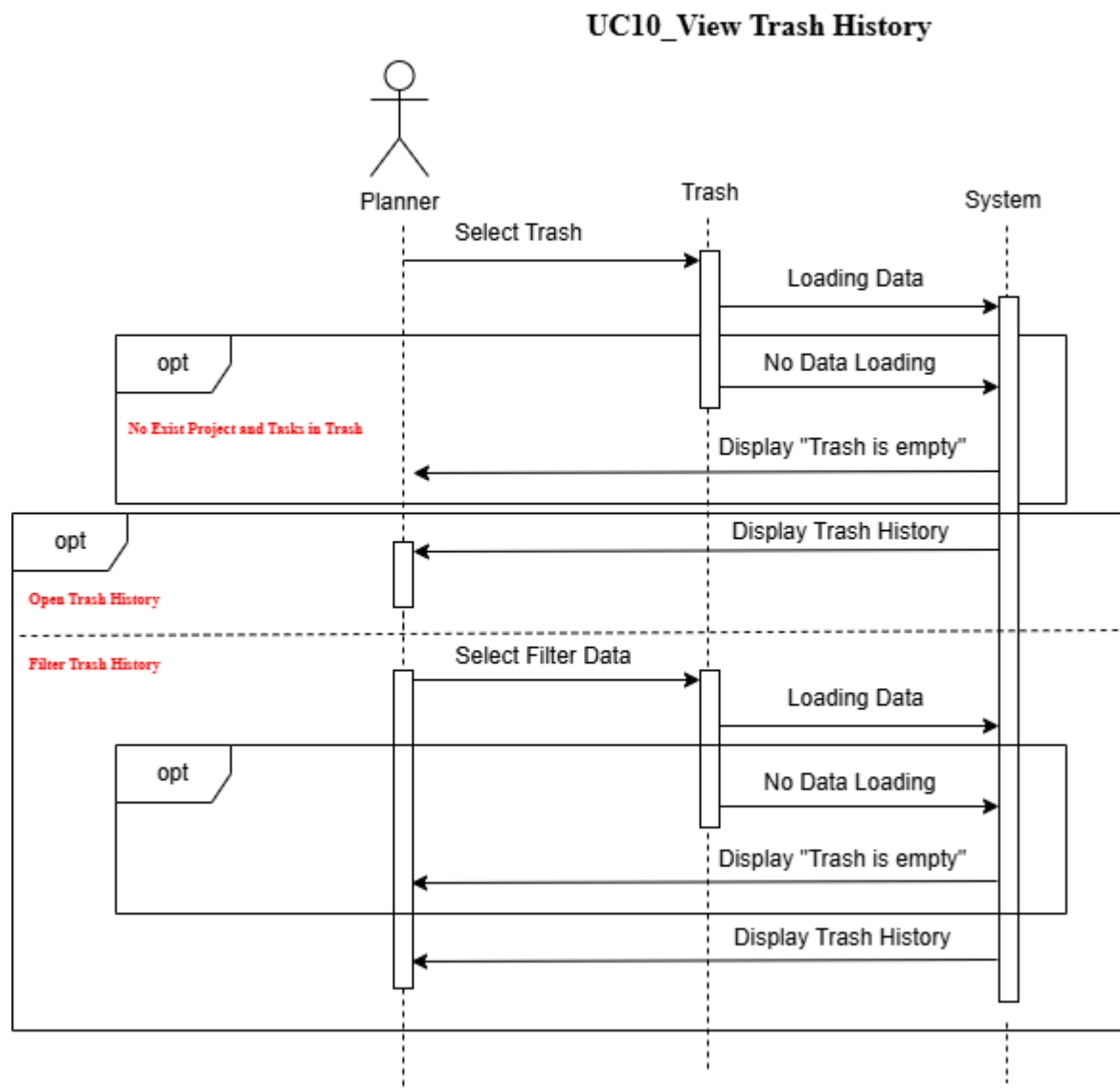


Figure 7 UC10_View Trash History (Sequence Diagram)

7. UC11_Using Real Time Tracker

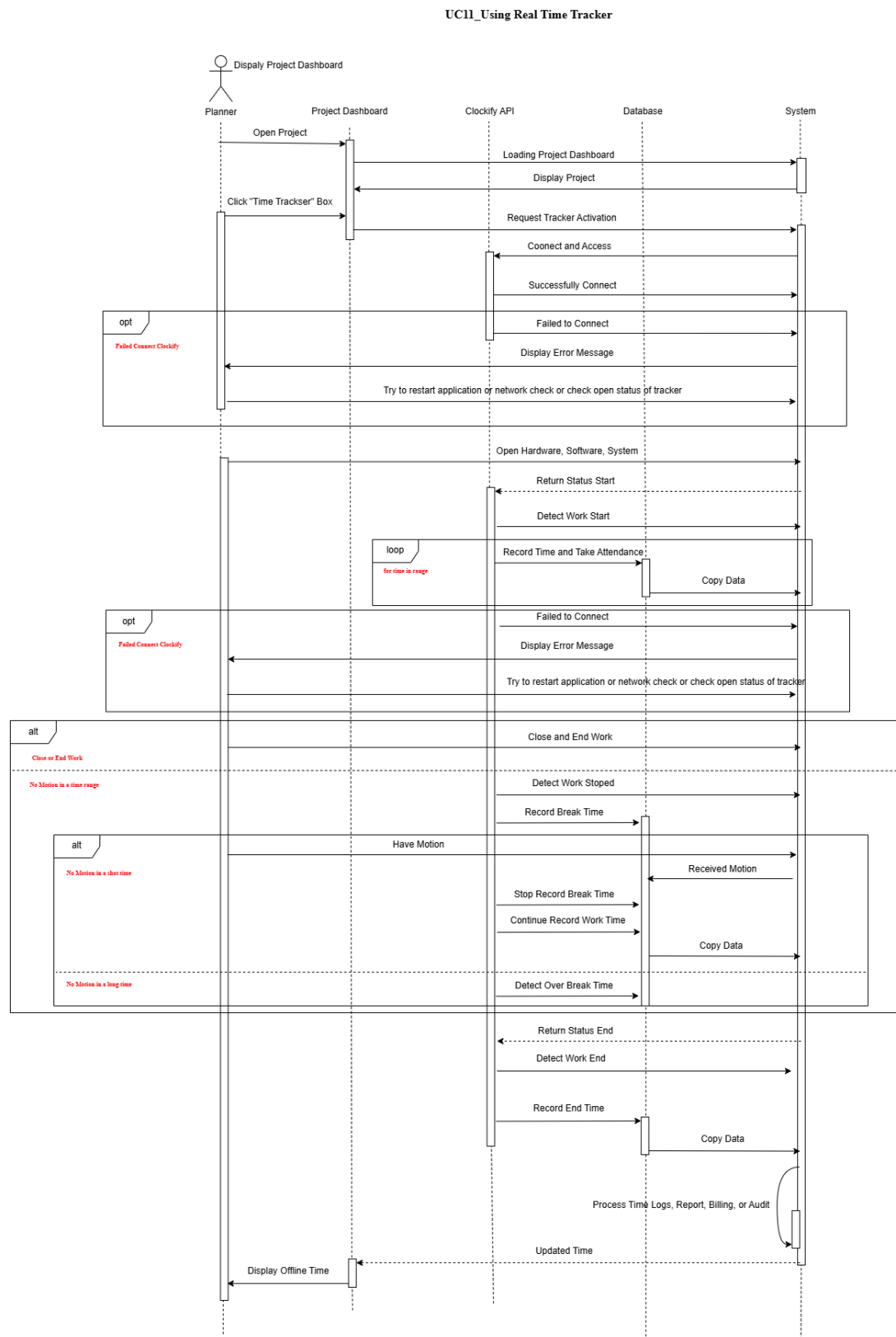


Figure 8 UC11_Using Real Time Tracker (Sequence Diagram)

8. UC12_View Visualize Graph

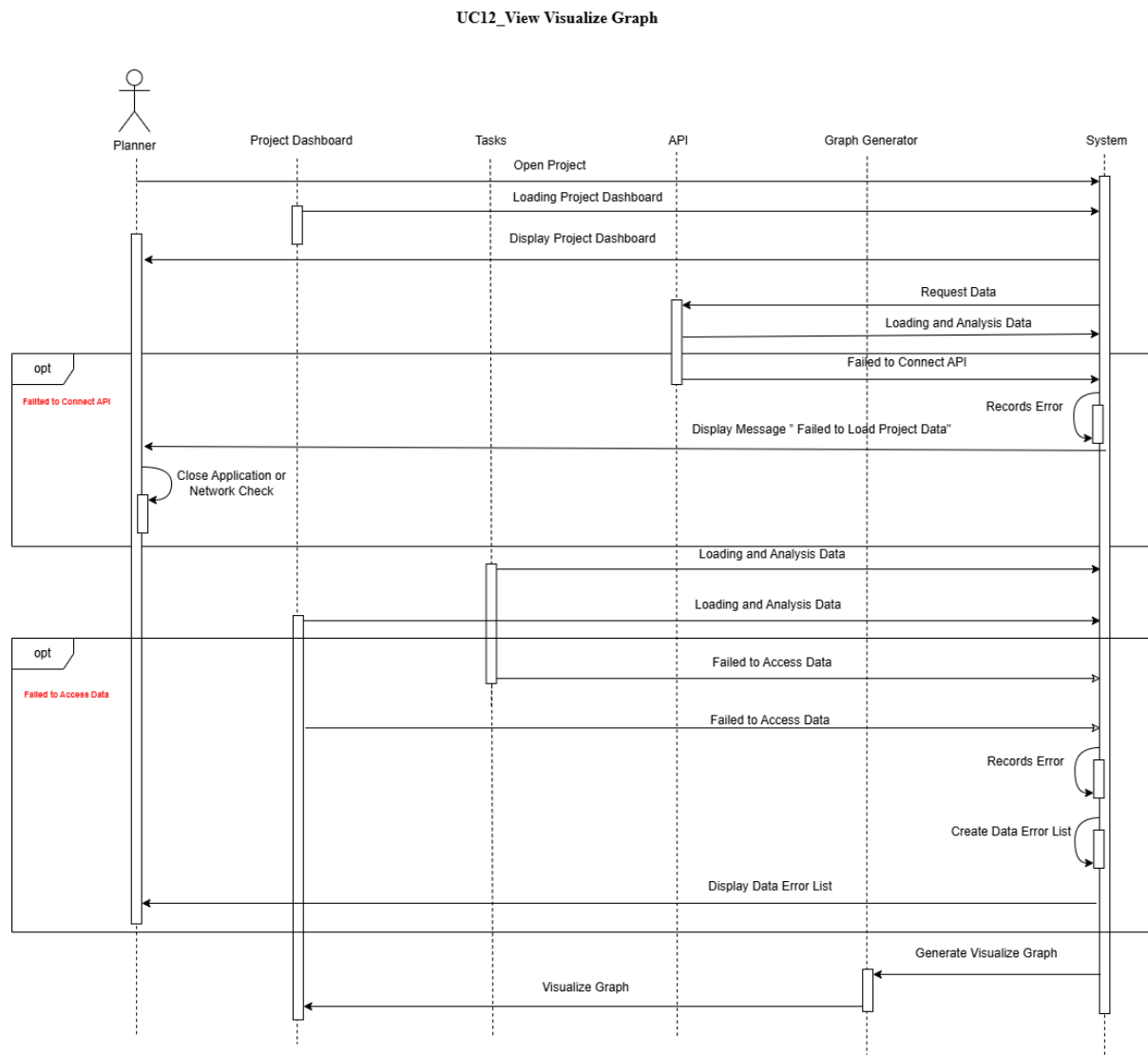


Figure 9 UC12_View Visualize Graph (Sequence Diagram)

1.3 State Machine Diagram

1. UC1_Create Project based with Templates or Tools

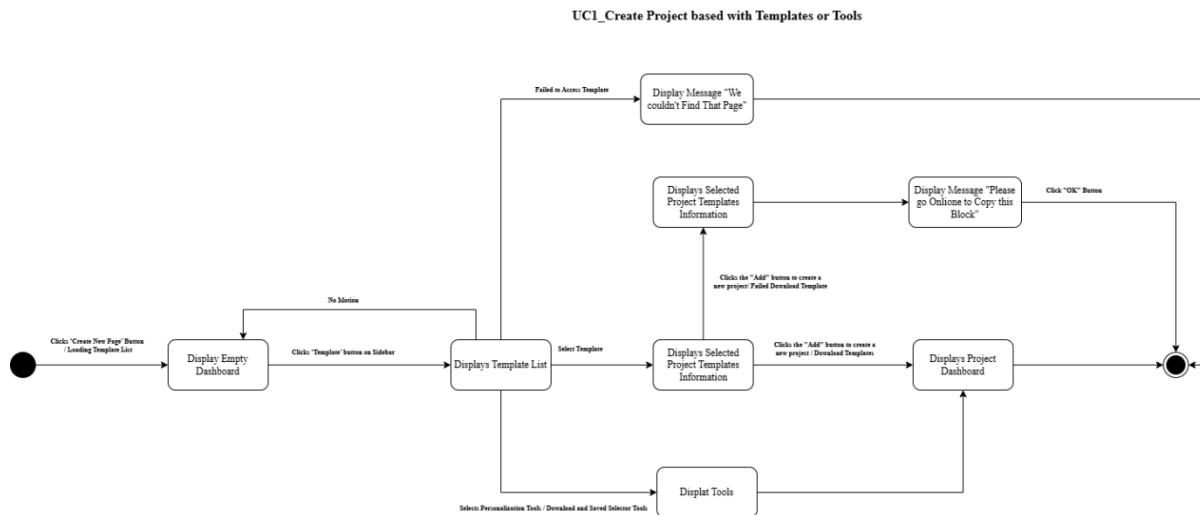


Figure 10 UC1_Create Project based with Templates or Tools (State Machine Diagram)

2. UC3_Manage Task

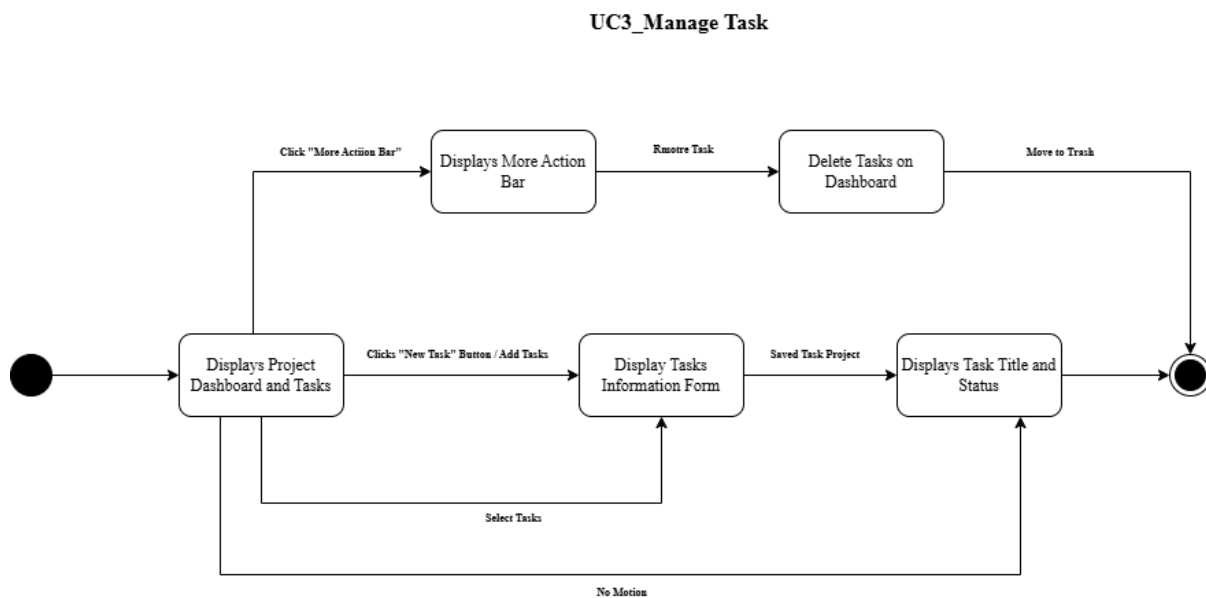


Figure 11 UC3_Manage Task (State Machine Diagram)

3. UC5_Open Relevant File with URL Link

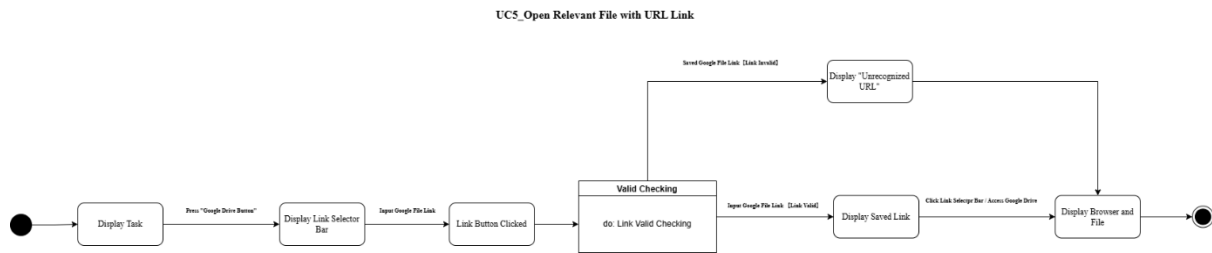


Figure 12 UC5_Open Relevant File with URL Link (State Machine Diagram)

4. UC7_Solve Questions Using Notion AI

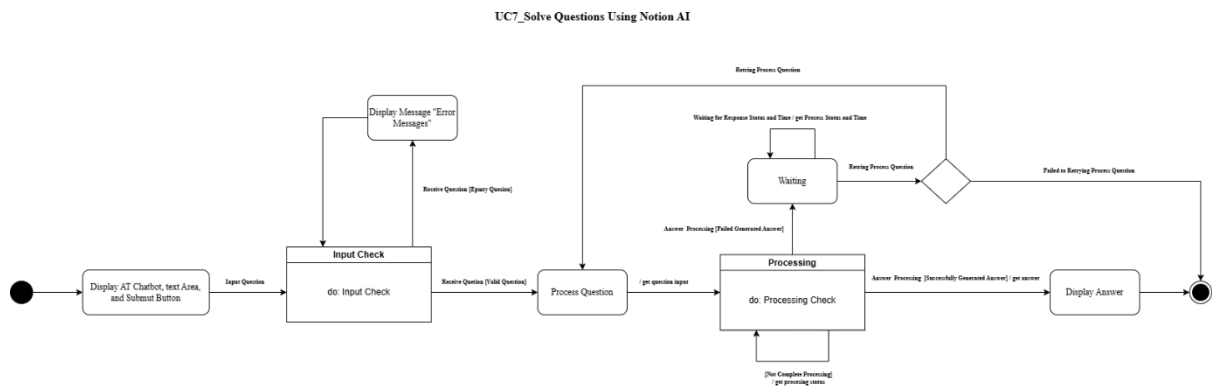


Figure 13 UC7_Solve Questions Using Notion AI (State Machine Diagram)

5. UC8_Searching Project and Task

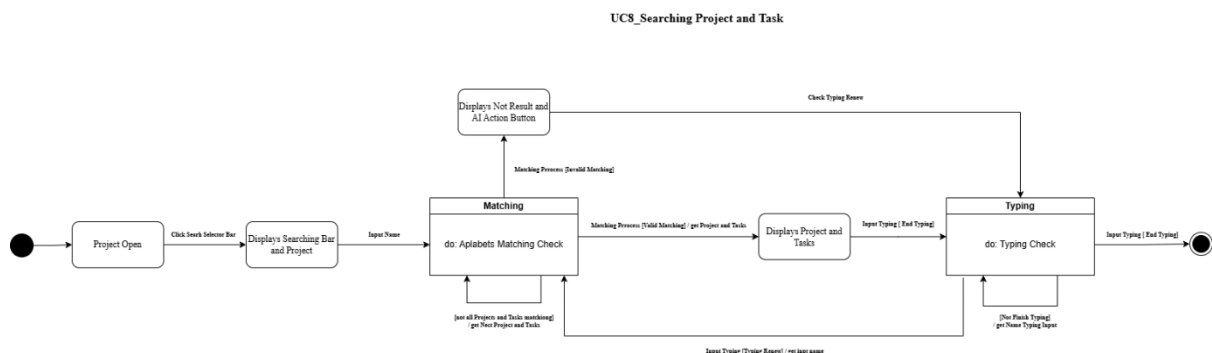


Figure 14 UC8_Searching Project and Task (State Machine Diagram)

6. UC10_View Trash History

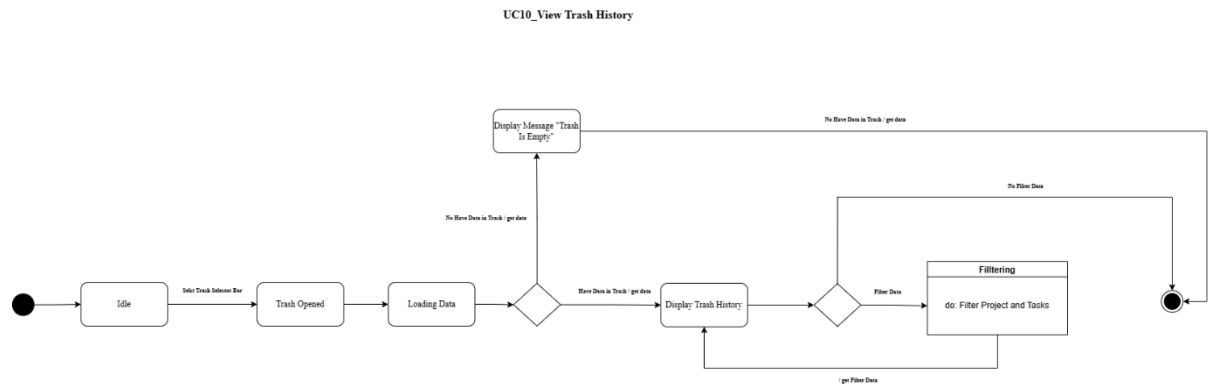


Figure 15 UC10_View Trash History (State Machine Diagram)

7. UC11_Using Real Time Tracker

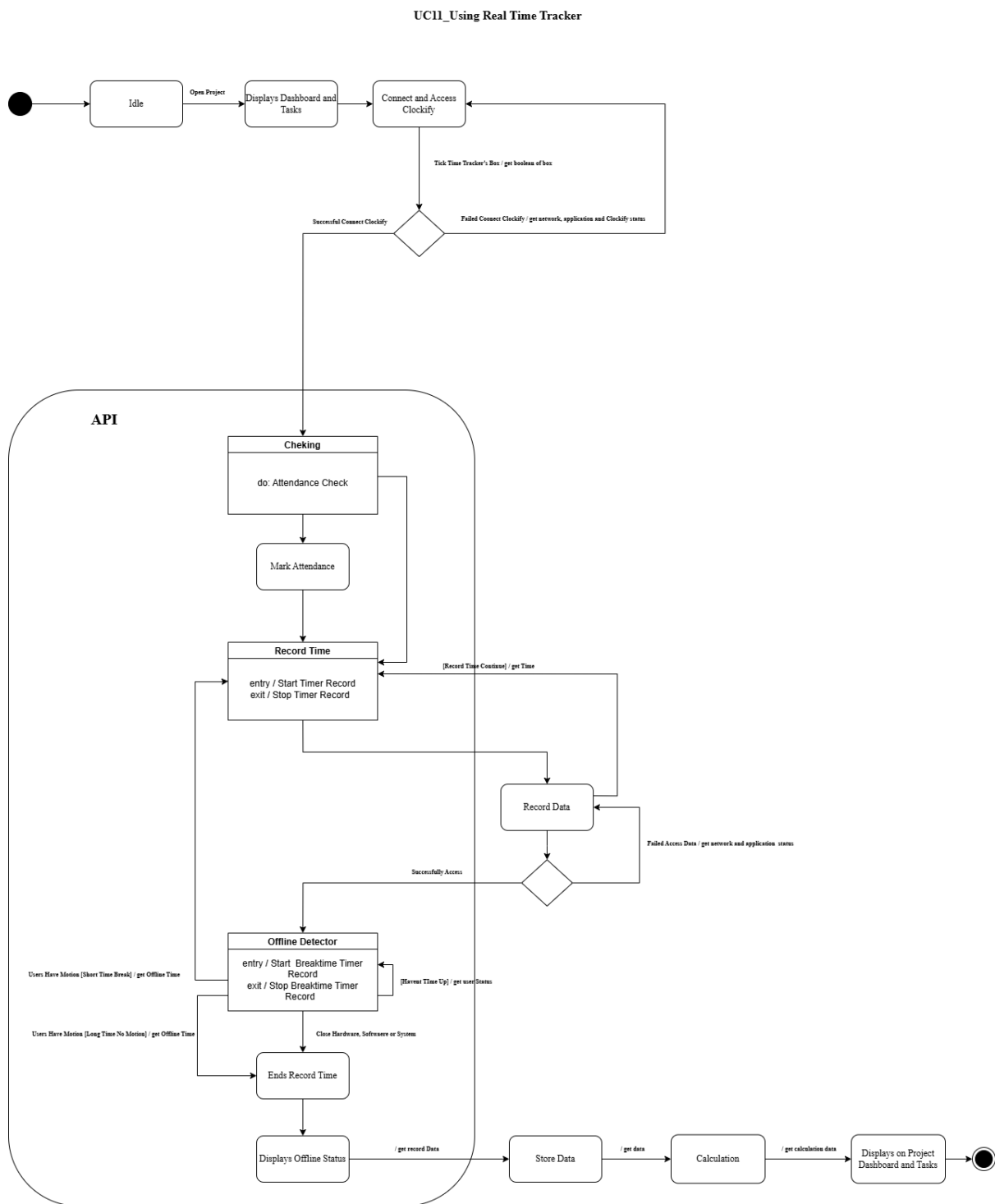


Figure 16 UC11_Using Real Time Tracker (State Machine Diagram)

8. UC12_View Visualize Graph

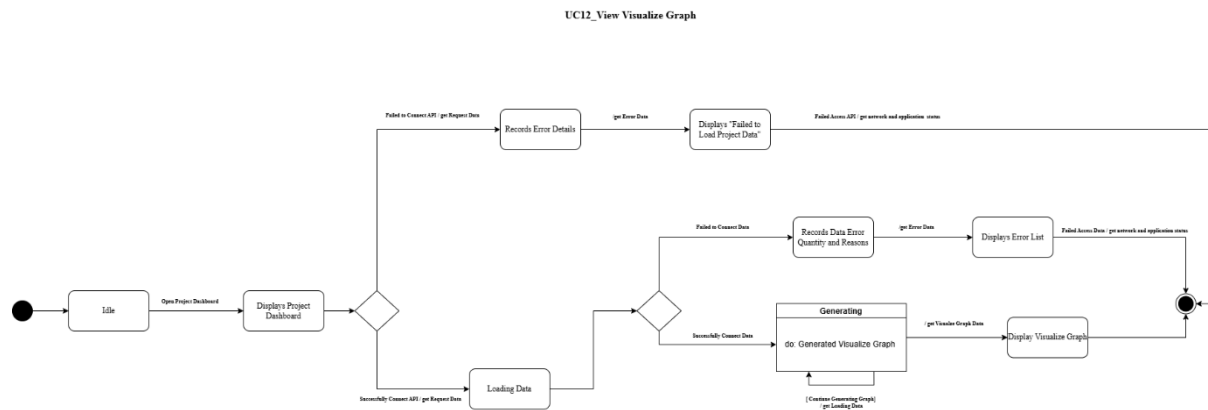


Figure 17 UC12_View Visualize Graph (State Machine Diagram)

2.0 Additional Diagram

2.1 Object Diagram

1. UC1_Create Project based with Templates or Tools

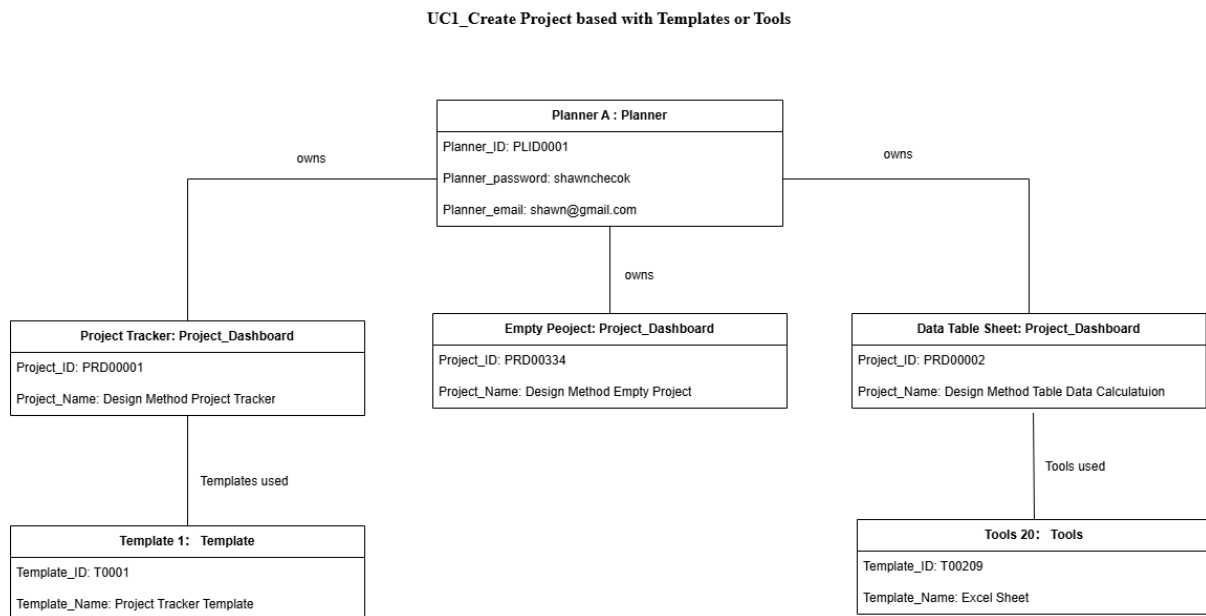


Figure 18 UC1_Create Project based with Templates or Tools (Objective Diagram)

2. UC12_View Visualize Graph

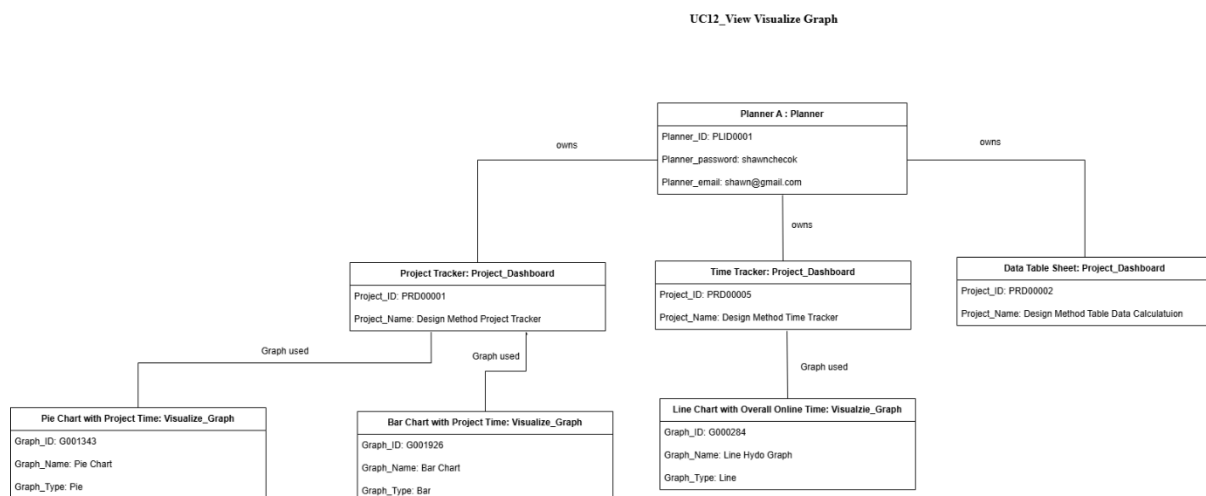


Figure 19 UC12_View Visualize Graph (Objective Diagram)

3.0 Conclusion

In conclusion, the system details of the Notion application were successfully to show out a few of the features and structures by using the design method. The functions in Notion have shown on creating the projects, Select Workspace, managing tasks, setting the times and dates, personalize notification, open relevant file with URL links, using AI to solve the questions, searching project dashboard, and view trash history. Apart from these functions, the improvement suggestion in the Notion extends two features are using real-time tracker and using visualize graph which have 12 features have to been design. The design UML diagram of the design method using are use case diagram, activity diagram, sequence diagram, state machine diagram, and object diagram.

In the use case diagram can visualize and document system functionality which can improve the communication and identifying system requirement. In activity diagram, the use case specification of each scenario have explained the details of postconditions, standard process, alternative flows, and exception flows. We also can view the overflow in the state of activity in this diagram to make sure the ideas of the system design are true. Besides, using the class diagram can measure the system development part can follow the object-oriented structure with using the class diagram. In the class diagram, I can learn the value, functions, methods, objects and their connection are suitable. In addition, the sequence diagram can learn the and visualize the interaction in the system which in the Object, message, states and time. Furthermore, the state machine diagram can provide a visual representation of states and transitions and make it easier to grasp the system such as the U11 using with time tracker. Lastly is the object diagram, it can provide a visual representation of specific instances of classes.

Although exception flows are described in the use cases, they are not clearly reflected in structural diagrams such as the activity diagram, or sequence diagrams. For the better system robustness and reliability, in the future improvements could include explicit modeling of error handling mechanisms. This could involve components like Error Logger, Validation Handler, or Retry Processor, which can manage failed API connections, invalid input, or data retrieval

errors. Representing these modules visually will provide a more complete and fault-tolerant system architecture.

In this assignment, I have learn how to draw UML diagram and the system designing ideas and it methods. Through this process, I gained a deeper understanding of system design concepts and the methods used to model software systems effectively. I also learned how to analyze user requirements and translate them into structured design components. This assignment enhanced my critical thinking, especially in mapping user interactions and backend processes into visual diagrams. Overall, it has strengthened my ability to approach system design systematically and communicate technical ideas clearly through visual models.

4.0 References

What is Class Diagram? (2024). Visual-Paradigm.com. <https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-class-diagram/>

What is Sequence Diagram? (2024). Visual-Paradigm.com. <https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-sequence-diagram/>

What is Object Diagram? (2024). Visual-Paradigm.com. <https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-object-diagram/>

What is Object Diagram? (2024). Visual-Paradigm.com. <https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-object-diagram/>

GeeksforGeeks. (2017, October 25). *Object Diagrams | Unified Modeling Language (UML)*. GeeksforGeeks. <https://www.geeksforgeeks.org/unified-modeling-language-uml-object-diagrams/>